

Neuroscience

Adaptive biasing of action-selective cortical build-up activity by choice history

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

[About eLife's process](#)

Reviewed preprint posted

10 May 2023 (this version)

Sent for peer review

21 March 2023

Posted to bioRxiv

3 February 2023

A. Braun , T.H. Donner 

Section Computational Cognitive Neuroscience, Department of Neurophysiology and Pathophysiology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany • Department of Psychiatry, Charité Universitätsmedizin, Berlin, Germany • Bernstein Center for Computational Neuroscience, Charité Universitätsmedizin, Berlin, Germany

 (https://en.wikipedia.org/wiki/Open_access) (<https://creativecommons.org/licenses/by/4.0/>)

Abstract

Decisions under uncertainty are often biased by the history of preceding sensory input, behavioral choices, or received outcomes. Behavioral studies of perceptual decisions suggest that such history-dependent biases affect the accumulation of evidence and can be adapted to the correlation structure of the sensory environment. Here, we systematically varied this correlation structure while human participants performed a canonical perceptual choice task. We tracked the trial-by-trial variations of history biases via behavioral modeling and of a neural signature of decision formation via magnetoencephalography (MEG). The history bias was flexibly adapted to the environment. The bias exerted a selective effect on the build-up (not baseline level) of action-selective motor cortical activity during decision formation, which added to the impact of the current stimulus. We conclude that the build-up of action plans in human motor cortical circuits is shaped by dynamic prior expectations that result from an adaptive interaction with the environment.

eLife assessment

This **valuable** work provides new insights into history-dependent biases in human perceptual decision-making. It provides **compelling** behavioral and MEG evidence that humans adapt their history-dependent to the correlation structure of uncertain sensory environments. Further neural data analyses would strengthen some of the findings, and the studied bias would be more accurately framed as a stimulus- or outcome-history bias than a choice-history bias because tested subjects are biased not by their previous choice, but by the previous feedback (indicating the category of the previous stimulus).

Introduction

Perceptual decisions made in the face of uncertain sensory evidence are often biased by previous stimuli, choices and choice outcomes ([Gold et al., 2008](#); [Busse et al., 2011](#); [de Lange](#)

et al., 2013; Akaishi et al., 2014; Fischer and Whitney, 2014; Fründ et al., 2014; Abrahamyan et al., 2016; Pape and Siegel, 2016; St. John-Saaltink et al., 2016; Fritzsche et al., 2017; Hwang et al., 2017; Urai et al., 2017; Braun et al., 2018). In most standard laboratory tasks, the environmental state (i.e., stimulus category) is uncorrelated across trials. In that context, such choice history biases tend to impair performance (Abrahamyan et al., 2016). However, when the environmental state exhibits some stability across trials, as is common for natural environments (Yu and Cohen, 2009; Glaze et al., 2015), choice history biases tend to improve performance (Braun et al., 2018). Indeed, a growing body of behavioral evidence shows that humans and other animals flexibly adapt their choice history bias to the correlation structure of the environment (Abrahamyan et al., 2016; Kim et al., 2017; Braun et al., 2018; Hermoso-Mendizabal et al., 2020).

How do adaptive choice history biases influence the formation of subsequent decisions? Prominent models conceptualize the decision formation as the accumulation of noisy sensory evidence into a decision variable (DV) that grows with time until a bound for one of the choice alternatives is crossed and that choice is triggered (Bogacz et al., 2006; Gold and Shadlen, 2007; Ratcliff and McKoon, 2008; Brunton et al., 2013; Ossmy et al., 2013). In such accumulator models, choice history biases may shift the starting point of the DV before evidence onset and/or bias the evidence accumulation per se. Behavioral modeling indicates that individual differences in the idiosyncratic history biases occurring in random environments are better explained by biases of evidence accumulation than by starting point biases (Urai et al., 2019). Such effects have not been assessed for adaptive biases in structured (stable or systematically alternating) environments. At the neural level, they should translate into a biased build-up rate (accumulation bias) of neural signatures of the DV, more so than an offset before decision formation (starting point).

Neural signals exhibiting signatures of the DV have been observed in parietal and frontal cortical areas involved in action planning in both primates (Shadlen and Kiani, 2013; Peixoto et al., 2021) and rodents (Hanks et al., 2015; Brody and Hanks, 2016). When choices are reported with hand movements, key properties of the DV are reflected in motor preparatory activity in (human and monkey) premotor and primary motor (M1) cortex. In human motor cortex, this selective motor preparatory activity is expressed in a suppression of ongoing beta-band oscillations contralateral to the upcoming hand movement, accompanied by an enhancement of gamma-band power (Crone, 1998a, 1998b; (Donner et al., 2009)) and likely spiking activity. While the origin of this beta-power suppression remains under study (Sherman et al., 2016; Little et al., 2019) we here use it as a functional marker of the DV that is encoded in local patterns of spiking activity: Like spiking activity (Shadlen and Kiani, 2013; Peixoto et al., 2021), only with inverted sign, the beta-band suppression (i) encodes the specific choice that will later be reported, (ii) gradually builds up during decision formation, with a rate that scales with evidence strength, and (iii), in reaction time tasks, converges on a common level just before action execution (Donner et al., 2009; O'Connell et al., 2012; Wyart et al., 2012; de Lange et al., 2013; Fischer et al., 2018; Wilming et al., 2020; Murphy et al., 2021).

We show that the sign and rate of the build-up of this selective DV-marker during evidence accumulation track a dynamic history bias, which is, in turn, adapted to varying environmental statistics. We combined a canonical decision-making task, discrimination of the net motion direction of dynamic random dot patterns (Gold and Shadlen, 2007), with systematic manipulations of the environmental statistics. Single-trial behavioral modeling uncovered the resulting history-dependent biases. Relating the model-inferred time-varying history bias to MEG measurements of the pre-trial baseline state and subsequent build-up rate of action-selective motor cortical population activity identified a neural signature of this adaptive bias in the latter, not the former.

Results

Human participants (N=38) performed a random dot motion (up vs. down) discrimination task with varying levels of motion strength spanning psychophysical threshold (Fig. 1A, Materials and Methods). We alternated the task, in pseudo-random order, between three different sensory environments with distinct repetition probabilities of stimulus categories (i.e., motion directions) across trials, referred to as Neutral, Repetitive, and Alternating, respectively (Fig. 1B). These three environments were characterized by approximately equal fractions of upward and downward motion stimuli, and they were presented in blocks of 99 trials each, separated by pauses (Materials and Methods). Participants were not informed about the existence of these different environments and received outcome feedback after each choice.

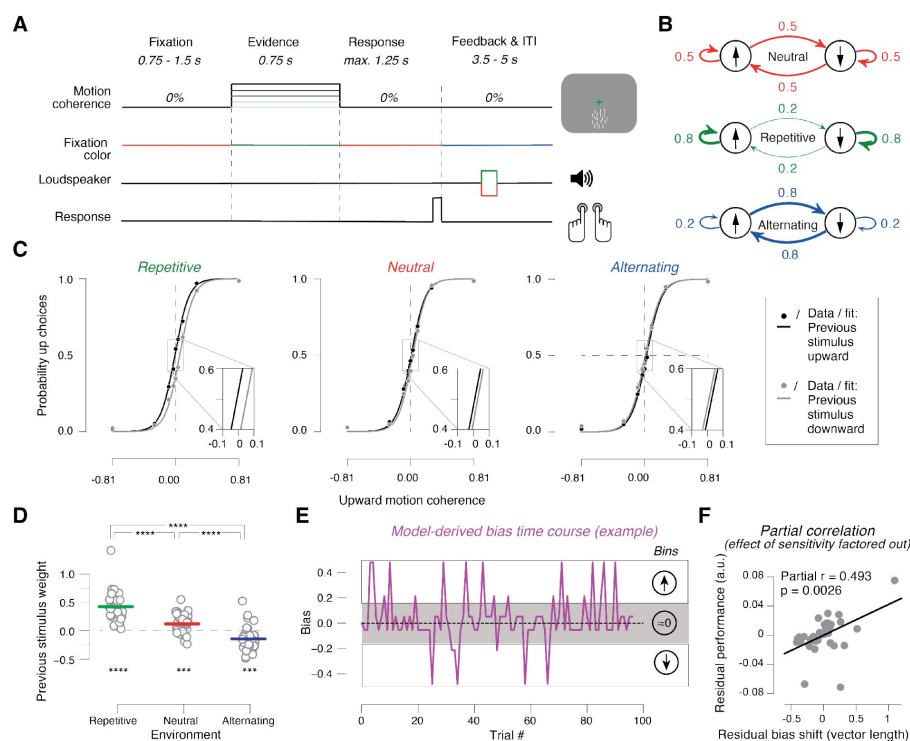


Figure 1.

Task and behavior in the different sensory environments.

A. Time course of events during a trial. Participants judged the net direction of motion of random dot kinematograms with varying levels of motion coherence and direction. 0% coherent motion was presented throughout the trial. Color switch of fixation cross indicated the onset of the decision interval with coherent motion (or 0% coherence on some trials). After 0.75 s, the color of the fixation cross switched back to red, to prompt the choice. After the button press or 1.25 s deadline, the fixation cross turned blue indicating the variable inter-trial interval with audi-

tory feedback. **B.** Manipulation of stimulus environments through variation of repetition probability of motion direction across trials. Repetition probability was 0.8 (Repetitive), 0.5 (Neutral), or 0.2 (Alternating). **C.** Psychometric functions conditioned on previous stimulus category (group average), for the three environments. Vertical lines, SEM (most are smaller than data points); insets, close-ups of the part in rectangle around 0% coherence indicating the systematic shift of history bias between the environments. **D.** Impact of previous stimulus categories on current choice for lag 1. Circles refer to values from individual participants. Lines refer to group means. *** $p < 0.001$, **** $p < 0.0001$ two-tailed permutation test. **E.** Single-trial history bias estimates for an example participant and block from the Neutral environment. Positive values correspond to a bias for choice 'up' and negative values correspond to a bias for choice 'down'. The magnitude indicates the strength of the bias. When binned into three bins of equal size, the low bin contains trials with a bias for choice 'down', the medium bin contains trials with a bias around zero and the high bin contains trials with a bias for choice 'up'. **F.** Bias adjustment improves performance. Partial regression between length of the vector of previous choice weights plotted against previous stimulus weights between Repetitive and Alternating in Fig. S2A and proportion of correct choices averaged across Repetitive and Alternating while factoring out the effect of sensitivity. Data points are the residuals from two separate regressions: length of vector difference on sensitivity (x-axis) and sensitivity on proportion correct (y-axis).

Adjustment of choice history biases to environmental context

We expected that the choice history biases would vary systematically between these different sensory environments, as observed in previous work (Abrahamyan et al., 2016; Braun et al., 2018; Hermoso-Mendizabal et al., 2020). Because the feedback after each trial disambiguated the previous stimulus category, we further expected that subjects might use that information for adjusting their history biases to the environment. We observed an indication of such an adjustment in their psychometric functions, when those were fit conditioned on the previous stimulus category (Fig. 1C). In all three environments, previous

category-dependent psychometric functions were shifted horizontally, indicative of a history bias (Repetitive: $t = 8.133$, $p < 10^{-4}$, Neutral: $t = 4.218$, $p = 0.0002$, Alternating: $t = -2.276$, $p = 0.0287$; two-tailed t tests). Critically, these shifts pointed in opposite directions for the two structured environments, with a strong tendency to repeat the previous category in Repetitive and a tendency to alternate the previous category in Alternating, which highlights the adaptive nature of the history biases (Fig. 1C). The previous stimulus category had no effect on perceptual sensitivity (history-dependent psychometric slopes: Repetitive: $t = -0.0397$, $p = 0.969$, $Bf_{10} = 0.175$, Neutral: $t = -0.623$, $p = 0.537$, $Bf_{10} = 0.209$, Alternating: $t = 0.094$, $p = 0.926$, $Bf_{10} = 0.175$; two-tailed t tests and Bayes factors).

We used a statistical model to quantify participants' history biases in a more comprehensive fashion and estimate single-trial bias time courses for the interrogation of MEG data in the subsequent sections. The model was fit separately to the choice behavior from each sensory environment and captured the history bias as a linear combination of the choices and stimulus categories from the recent trials (Materials and Methods). We used a cross-validation procedure to select the best-fitting model order (i.e., number of previous trials contributing to the bias), separately for each individual and each environment (Fig. S1) and applied this model to independent data in order to estimate subjects' history weights (Fig. 1D; Fig. S2) and construct bias time courses (Fig. 1E). The analyses presented in the following included only those subjects (all but two), which showed a best-fitting lag larger than 0 in at least one of the two biased environments (commonly Repetitive, Fig. S1).

The estimated model parameters (cross-validated regression weights, Fig. 1D, Fig S2) showed a pattern in line with the psychometric function shifts in Fig. 1C. In Fig. 1D, positive regression weights for the previous stimulus category indicated a tendency for subjects to repeat (in their choice) the previously shown stimulus category. Likewise, negative weights indicated a tendency to alternate the choice relative to the previous stimulus category. The impact of the previous trial stimulus category on current choice was different from zero in all three environments, including Neutral (Fig. 1D; Repetitive: $p < 10^{-5}$; Alternating: $p = 0.0003$; Neutral: $p = 0.0002$; two-tailed permutation tests). But critically, in both biased environments, this impact was different from Neutral and shifted in opposite directions, indicating a tendency to repeat the previous stimulus category in Repetitive and vice versa in Alternating (Fig. 1D; Repetitive vs. Neutral: $p < 10^{-5}$; Alternating vs. Neutral: $p < 10^{-5}$; two-tailed permutation tests).

The impact of the previous choice on the current choice tended to be overall weaker, more idiosyncratic, and less systematically related to the sensory environment than the impact of the previous stimulus (Fig. S2A). Indeed, the shift in stimulus weights between each biased condition and Neutral was significantly larger than the corresponding shift in choice weights (Repetitive $p = 0.0002$, Alternating: $p = 0.0026$; two-tailed permutation test). There was little contribution of the stimulus categories from trials further back in time (Figure S2B). Overall, the pattern of model parameters is consistent with our expectation that participants' adjustment of their history biases would be governed by the previous stimulus category, which was disambiguated through the trial-by-trial feedback.

Indeed, the individual degree of history bias adjustment made a significant contribution to individual performance (Fig. 1F). We computed an individual measure of bias adjustment from the weights of both previous stimulus and choice (Materials and Methods) and used this to predict participants' overall task performance in the structured environments (proportion of correct choices collapsed across Repetitive and Alternating). As expected, individual performance also strongly depended on participants' sensitivity to the current evidence (i.e., slope of the psychometric function). We, therefore, used partial regression to quantify the unique contribution of each factor (history bias adjustment and evidence sensitivity) on performance. Both factors uniquely predicted performance (sensitivity: $r = 0.798$, $p < 0.0001$; bias adjustment: $r = 0.493$, $p = 0.0026$; Pearson correlation), with a clear

effect of the adjustment of history biases (Fig. 1F). The same was true when we used the individual weights of the previous stimulus for performance prediction, separately for the two biased environments, but not the Neutral environment (Fig. S3).

Large-scale cortical dynamics of task processing

The behavioral results reported above indicate that participants adjusted their history bias to the environmental statistics, which, in turn, boosted their performance. How did these (partly) adaptive history biases affect the formation of subsequent decisions, more specifically: the dynamics of the underlying DV in the brain? Our concurrent collection of whole-brain MEG data during this task enabled us to address this question. We combined source reconstruction with established anatomical atlases and spectral analysis to characterize the large-scale cortical dynamics involved in our task (Wilming et al., 2020; Murphy et al., 2021).

We first identified established neural signatures of the motion stimulus strength and of selective action planning during decision formation in our current data. In line with previous work (Siegel et al., 2007), gamma-band power (~60-100 Hz) in visual cortical areas was enhanced while low frequency power (<30 Hz) was suppressed relative to baseline during motion viewing (Fig. 2A); both components of the visual responses scaled with motion coherence, predominantly in dorsal visual cortical areas V3A/B, IPS0-3, and the MT+ complex (Fig. 2B). Concomitantly with these responses to visual motion, activity lateralization predicting the subsequent choice (left vs. right button) built up in downstream (anatomically more anterior) parietal and motor cortical areas (Fig. 2C). Again in line with previous work (Donner et al., 2009; de Lange et al., 2013; Pape and Siegel, 2016; Wilming et al., 2020; Murphy et al., 2021), this action-selective activity build-up was a suppression of beta-band (12-36 Hz) power contra- vs. ipsilateral to the upcoming movement, and robustly expressed in the M1 hand area (Fig. 2C and Fig. 4A). This signal, referred to as ‘motor beta lateralization’ in the following, has been shown to exhibit hallmark signatures of the DV in different experimental task contexts (Siegel et al., 2011; Murphy et al., 2021; O’Connell and Kelly, 2021).

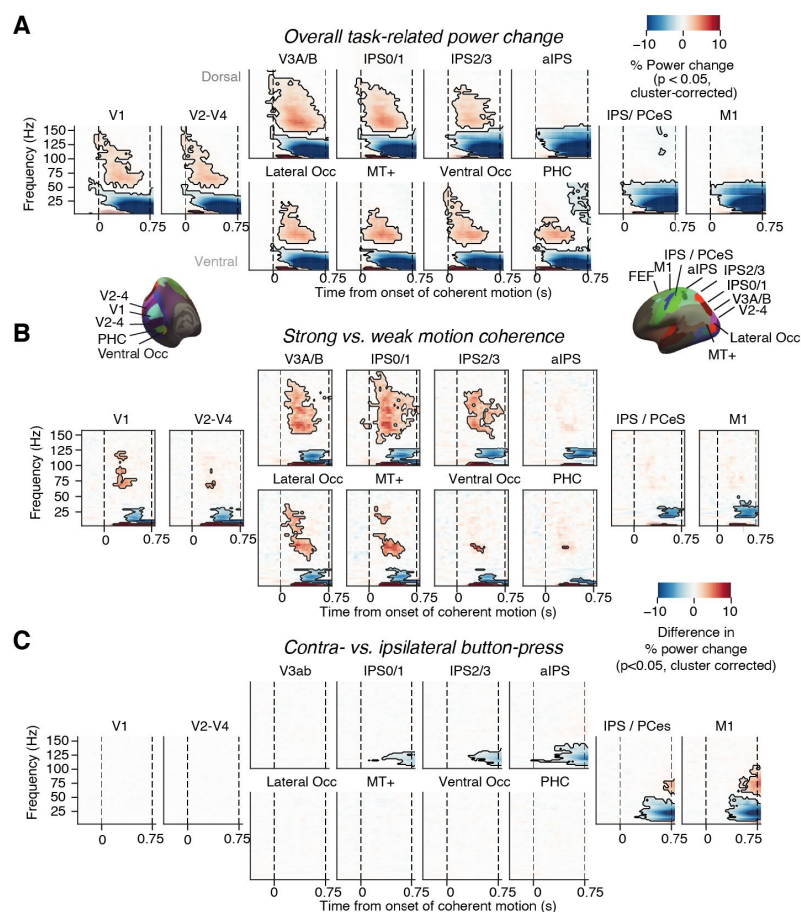


Figure 2.

Neural signatures of stimulus processing and action planning across the cortical visuo-motor pathway.

A. Overall task-related power change (average across hemispheres). Increase in visual gamma-band response and decrease in alpha- and low-beta-band power in visual cortex during presentation of coherently moving dots. **B.** Motion coherence specific sensory response. Difference in time-frequency response between high (0.81%) and 0% motion coherence (average across hemispheres). Increase in visual gamma-band power and decrease in alpha- and low-beta-band power scale with motion coherence of stimulus. **C.** Time-frequency representation of action-selective power lateralization contralateral vs. ipsilateral to upcoming button press. All signals are expressed as percentage of power change relative to the pre-trial baseline. Dashed vertical lines, onset and offset of coherent motion. Saturation, significant time-frequency clusters ($p < 0.05$, two-tailed cluster-based permutation test).

In sum, we replicated well-established signatures of visual motion processing and action-selective motor preparation in our current MEG data. We next studied two fundamental aspects of the motor preparation signal, its baseline level at the start of the decision process and its build-up during decision formation, to comprehensively assess the impact of the adaptive history bias on the dynamics of this neural marker of the DV.

No consistent modulation of baseline state of action-selective activity by environmental context and trial history

Previous human MEG work indicates that the motor cortical baseline beta-power state is flipped relative to its state just before the previous choice, a phenomenon referred to as ‘beta rebound’ (Pfurtscheller et al., 1996; Pape and Siegel, 2016; Urai and Donner, 2022) that was also evident in our data (Fig. 3A, collapsed across all three environments). Recent MEG studies of human perceptual decision-making have linked this phenomenon to either overt choice alternation (Pape and Siegel, 2016) or alternating starting points inferred from drift diffusion model fits (Urai and Donner, 2022). We, therefore, wondered if and how the baseline level of motor beta lateralization depended on the different sensory environments or on the history bias in specific trials.

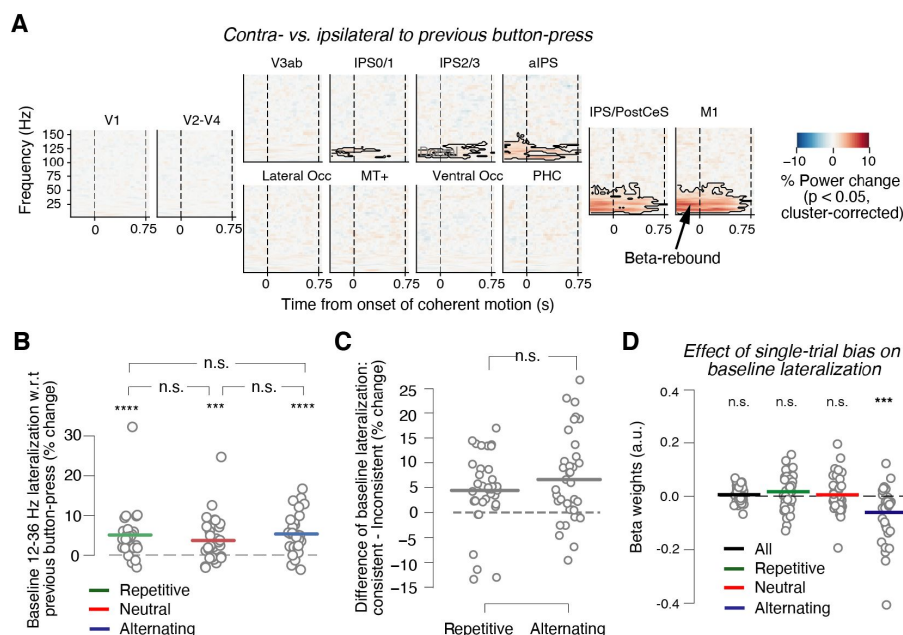


Figure 3.

Baseline state of motor cortex reflects previous choice, but not consistently context or history bias.

A. Spill-over of action-selective beta-power rebound from previous into current trial. Time-frequency representation of power lateralization contra- vs. ipsilateral to the previous button-press, expressed as percentage power change from baseline. Enhanced beta-band power contra- vs. ipsilateral to the previous button-press in motor cortices. Dashed vertical lines mark the onset and offset

of coherent motion. Saturation, significant time-frequency clusters ($p < 0.05$), two-tailed cluster-based permutation test across participants. **B.** Impact of sensory environment on overall baseline state of beta lateralization (350 ms to 100 ms before stimulus onset) contra- vs. ipsilateral to previous button-press. **C.** Difference in baseline beta lateralization (relative to final button-press) for trials with choice consistent vs. inconsistent with previous stimulus, sorted by sensory environment. **D.** Impact of single-trial history bias on amplitude of M1 beta lateralization (relative to up-coding hand) during baseline interval (from 350 to 100 ms before evidence onset). *** $p < 0.001$, **** $p < 0.0001$ (two-tailed permutation test).

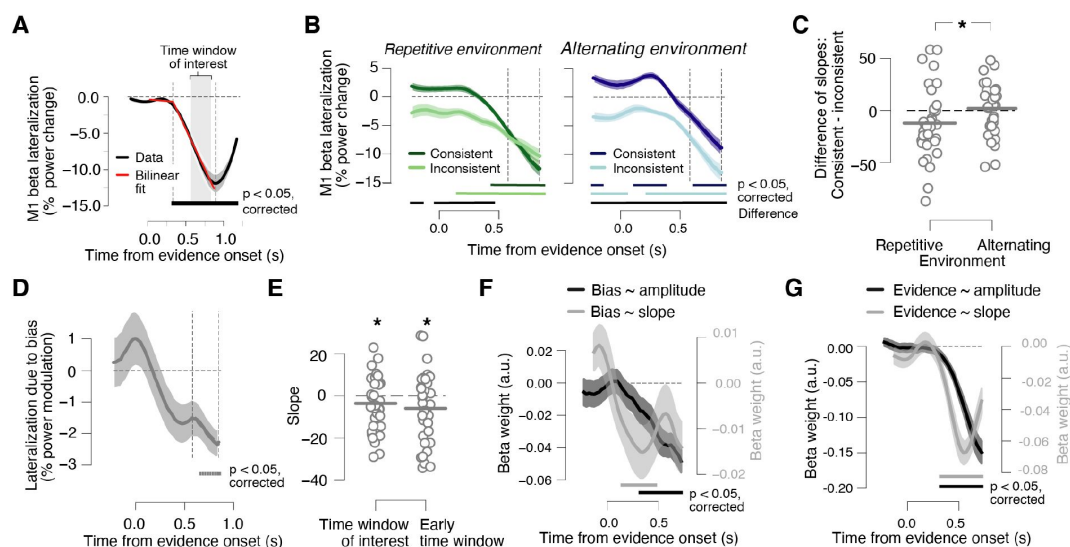


Figure 4.

Adaptive biasing of action-selective build-up activity in M1.

A. Time course of action-selective beta power (12–36 Hz) lateralization in the M1 hand area, contralateral vs. ipsilateral to upcoming button press, collapsed across trials (black line). Red line, bilinear fit. Gray box, time window (0.58 s to 0.8475 s from evidence onset) used to quantify the (rate of) build-up of power lateralization in panels B–E (vertical dashed lines in B and D). The window was defined to start 250 ms after the intersection point of bilinear fit and end 50 ms before the minimum of power lateralization, chosen so as to cover the interval containing ramping activity in the majority of trials. **B.** Same as A, but now split by sensory environment (Repetitive vs. Alternating) and by consistency of the upcoming behavioral choice with the stimulus category from previous trial. Dark colors, choice on current trial = stimulus category on previous trial; bright colors, choice on current trial = – stimulus category on previous trial. **C.** Difference in slopes between consistent and inconsistent conditions from B in time window of interest (see A), separately for Repetitive and Alternating environments. **D.** Component of action-selective lateralization governed by single-trial bias, irrespective of upcoming behavioral choice and pooled across sensory environments (see main text for details). **E.** Slope estimates for neural bias measures from panel D. Left, time window from panel A. Right, early time window derived from single-trial regression in panel F. **F.** Time-variant impact of single-trial history bias on amplitude (black) and slope (gray) of M1 beta lateralization (relative to up-coding hand). **G.** Same as F but for impact of signed stimulus strength. Shaded areas, SEM. Bars, $p < 0.05$ (two-tailed cluster-based permutation test) across participants. * $p < 0.05$ (one-tailed paired permutation test).

If this baseline lateralization state ‘inherited from’ the previous choice was involved in mediating the effect of the adaptive bias on choice, it might be expected to be reduced in Repetitive vs. Alternating environments, thus reducing subjects’ tendency to alternate in the former. Instead, the beta rebound effect (i.e., increased power contralateral vs. ipsilateral to previous choice) was about equally strong in all three environments (Fig. 3B). We found no evidence for its modulation by sensory environment (Repetitive vs. Neutral: $p = 0.1132$; Repetitive vs. Alternating: $p = 0.8167$; Neutral vs. Alternating: $p = 0.1067$; all two-sided permutation tests)

We then related the baseline M1 beta lateralization to the adaptive history bias. In a first model-independent approach, we conditioned the baseline motor-beta lateralization contra- vs. ipsilateral to the final button-press on the consistency of the final choice with the previous stimulus category, separately for Repetitive and Alternating (Fig. 3C). If the adaptive history bias shifted the motor baseline state, one might expect distinct consistency-dependent modulations of the baseline state in the different environments. For the Repetitive environment, one might expect a smaller baseline beta lateralization on trials ending with choices that were consistent with the previous stimulus category (i.e., reflecting a larger bias towards stimulus repetition) than on trials ending with a choice inconsistent with the previous category. For the Alternating environment, one might expect the opposite pattern. Instead, we found no difference of the baseline beta lateralization for consistent vs. inconsistent between Repetitive and Alternating ($p = 0.0972$, one-tailed permutation test).

The previous analyses did not capture the model-inferred bias on a trial-by-trial basis. We next adapted a single-trial regression procedure from recent monkey physiology work (Mochol et al., 2021) to relate the time-varying history bias to neural data in order to test if this bias modulated the baseline motor beta lateralization on a trial-by-trial basis. We used each individual's time course of single-trial history bias estimated through the behavioral model (positive values for bias toward upward, Fig. 1E) as predictors for the single-trial motor beta lateralization, whereby lateralization was assessed relative to the hand coding for up-choices in a given block (Materials and Methods). This procedure took the impact of both previous stimuli as well as previous choices into account, estimated with an individually optimized number of lags. Thus, the single-trial bias estimates were largely independent of assumptions about the sources of the single-trial bias (stimuli, choices, lags). However, because the model was fit and applied separately to data from different environments, the resulting time course of single-trial bias estimates captured the context-dependent, adaptive bias components described in the preceding section. An involvement of the motor baseline state in the implementation of history bias predicts a stronger baseline beta-suppression contralateral to the hand favored by the bias, with a magnitude that scales with the strength of the bias. In other words, this scenario predicts significant negative beta coefficients, regardless of the environment.

We found no such effect when the analysis was run across all three environments (Fig. 3D, 'All'), again inconsistent with the notion of a generally bias-encoding neural signal. We did find an effect of the single-trial bias on the baseline beta lateralization state in the Alternating environment when analyzed selectively (Fig. 3D; $p = 0.0003$, two-tailed permutation test). Such an effect was, however, not present for either of the other two environments (Fig. 3D; All: $p = 0.2838$; Repetitive: $p = 0.1343$; Neutral: $p = 0.6571$; two-tailed permutation tests). Overall, the results of our analyses of the baseline beta lateralization suggest the beta rebound from the previous trial may help promote choice alternation when performing in an Alternating context, but does not generally encode adaptive choice history biases.

Adaptive history bias shapes the build-up of action-selective motor cortical activity

The analyses from the previous section assessed the dependence of the starting point (i.e., baseline level) of a neural DV-proxy on environmental context and adaptive history bias. Behavioral modeling has shown that idiosyncratic choice history biases in a variety of tasks in random environments are accounted for by history-dependent biases in the build-up (i.e., drift) of the DV, rather than in its starting point (Urai et al., 2019). We, therefore, next asked whether the adaptive history biases identified here might shape the build-up rate of our neural proxy of the DV during decision formation.

Again, in a model-independent approach, we conditioned the time courses of the motor-beta lateralization on the consistency of the current choice with the previous stimulus category, separately for the three environments. If the history bias shaped the build-up of action-selective activity in an adaptive fashion, one might expect opposite patterns of ramping slopes in Repetitive versus Alternating environments: In the Repetitive environment, slopes might be steeper on trials ending with a choice consistent with the previous stimulus category (reflecting the bias towards repetition) and the opposite for the Alternating environment. The data were in line with these predictions (Fig. 4B, C). We quantified this effect as the difference in the ramping slopes between consistent and inconsistent, assessed separately for Repetitive and Alternating (Fig. 4C). The slopes were estimated with linear regression for an interval that exhibited clear linear ramping in the average motor beta lateralization across all trials (grey-shaded in Fig. 4A, Materials and Methods). Consistent vs. inconsistent slope differences had opposite sign on average, and differed between Repetitive and Alternating environments when tested across the group ($p = 0.0285$, one-tailed permutation test). This difference indicates an adjustment of the ramping of action-selective motor cortical activity to the environmental context. Note that before evidence onset, the motor-beta lateralization was larger when the current choice was consistent than inconsistent with the previous stimulus category in both biased environments due to the motor beta-rebound from the previous response (see also Figure 3C). We obtained qualitatively identical results as in Fig. 4B, C when first removing (using linear regression) the beta-rebound from the baseline lateralization levels (data not shown).

These model-independent results were corroborated by two complementary approaches that again exploited our model-inferred single-trial bias estimates. In one of those approaches, we grouped the single-trial bias estimates into three equally-spaced bins, with two bins containing strong biases of opposite direction (up vs. down) and the middle bin containing trials with little bias (Fig. 1E). We used the ‘up’ and ‘down’ bins to visualize the impact of the history bias on the ramping of the neural DV, by computing the time course of beta lateralization contra- vs. ipsilateral to the button-press for the direction of the bin-wise bias (Materials and Methods). The behavioral choice was, by definition, correlated with both, the single-trial bias as well as the action-selective motor beta lateralization, (Figs. 2C and 4A). This could yield a correlation between bias and motor cortical lateralization even in the absence of any direct effect of bias on motor beta lateralization. To isolate a genuine effect of the bias on our neural DV, we subsampled the data from the up and down bias bins to yield an equal number of upward and downward choices within each bin (Materials and Methods). For each bin, we then computed the time course of beta lateralization relative to the button press for the bias and collapsed the resulting time courses across bins. This procedure isolated the impact of the model-inferred history bias on the neural DV, independent of the choice. The resulting time course ramped into the direction of the single-trial bias, reaching statistical significance at about 700 ms after motion onset, before the end of the decision interval (Fig. 4D). We used linear regression to estimate the slope of the ramp, again focusing on the time window from Fig. 4A. As expected, the slope was smaller than zero ($p = 0.048$; one-tailed permutation test against zero; Fig. 4E, left), indicating that the time-varying history bias contributed to the build-up of action-selective motor cortical activity during decision formation.

Second, we again fit a single-trial regression model, now to simultaneously quantify the impact of the history bias and current evidence on the dynamics of the neural DV in a time-variant fashion. We ran two separate regression models, one on the amplitude of motor beta lateralization for a range of time windows, the other on the slope of motor beta lateralization, assessed locally in time for the same time windows; in both cases, lateralization was again assessed relative to the hand coding for up-choices in a block (Materials and Methods). An impact of the adaptive history bias on the ramping of motor

cortical activity would predict a specific effect of the history bias, over and above the effect of the sensory evidence, on both read-out measures, in particular on the ramping slopes. Specifically, it predicts negative beta weights, reflecting steeper downward slope (i.e., stronger suppression) for stronger biases.

We found a clear and expected effect of current sensory evidence on motor beta lateralization, with a steeper downward slope for stronger evidence (Fig. 4G). Critically, and in line with our hypothesis, the same was true for the effect of the history bias: a stronger bias produced a stronger and steeper motor beta lateralization towards the direction of the bias (Fig. 4F). The bias effect on the lateralization amplitude reached significance during the decision interval (from about 320 to about 720 ms after motion onset; Fig. 4F, black line), and the corresponding impact on the ramping slope was significant even earlier, during the first half of the decision interval (starting at about 150 ms after the motion onset; Fig. 4F, grey line). Combined, these two effects indicate that a strong bias on a given trial constituted an early force on the M1 ramping dynamics, pushing the signal into the direction of the bias even before the current evidence exerted its effect (compare with gray lines in Fig. 4G); the M1 lateralization amplitude later during the decision interval reflected the bias more strongly on trials, for which the bias was strong than those, for which the bias was weak.

Our analysis of the ramping slopes in Fig. 4E (left) estimated the slope for a longer (and later) time interval than the one, for which the single-trial regression in Fig. 4F yielded significant slope effects. We, thus, repeated the above analysis also for the earlier time window derived from the single-trial regression results Fig. 4E (right). Also, for this window did we find a robust effect of the bias estimate on the ramping slope ($p = 0.0158$; one-tailed permutation test against zero). Taken together, model-independent and model-based analyses provided convergent evidence for the dependence of action-selective cortical ramping activity during decision formation on the time-varying, context-dependent history biases.

Discussion

It has long been known that the history of preceding choices and stimuli biases perceptual judgments of the current stimulus (Fernberger, 1920). Recent behavioral modeling showed that at least part of such choice history biases reflect time-varying expectations that are flexibly adapted to the environmental structure (Abrahamyan et al., 2016; Braun et al., 2018; Hermoso-Mendizabal et al., 2020). Such dynamics, largely ignored in standard neurophysiological studies of perceptual decision-making, may be a key driver of sensory-guided behavior in ecological settings (Mobbs et al., 2018). How adaptive expectations shape the neural dynamics underlying decision-making has remained unknown. Here, we addressed this issue by combining a standard task from the neurophysiology of decision-making (Gold and Shadlen, 2007; Siegel et al., 2011) with systematic manipulations of the environmental stability as well as single-trial, model-based MEG-assessment of cortical decision dynamics. This revealed that the history-dependent, dynamic expectations boosted participants' behavioral performance and selectively altered the build-up sign and rate, not (consistently) the pre-trial baseline level, of an established neurophysiological proxy of the DV: action-selective preparatory population activity in their motor cortex.

While participants' history biases in a random environment (i.e. uncorrelated stimulus sequences) were largely idiosyncratic, as widely observed (Akaishi et al., 2014; Urai et al., 2019; Urai and Donner, 2022), we found that one component of these biases lawfully shifted between stable (frequent category repetitions) and systematically alternating environments and improved participants' performance. It is instructive to compare this adjustment of history bias with the one observed in a previous study using a similar manipulation of

environmental statistics (Braun et al., 2018). In that previous study, participants did not receive outcome feedback and thus remained uncertain about the category of the previous stimulus. Correspondingly, the history bias adjustment was evident in the impact of their previous choices (rather than previous stimulus categories), and most strongly of those made with high confidence (i.e., correct and fast). By contrast, in the current study, participants could deterministically infer the true category of the previous stimulus from the feedback. Correspondingly, we found that their history bias adjustment to the different environments was now governed by the previous stimulus category. Together, the findings from both studies support the notion that human subjects can use different types of internal signals to build up history-dependent expectations in an adaptive fashion.

The observation of effective behavioral adjustment to differentially structured environments in participants' steady-state behavior raises the question of how (and how quickly) they learned the different environmental structures. Participants received no information about this experimental manipulation during task performance. Informal debriefings after the experiment in a subsample indicated that most did not become aware of the environmental manipulation, despite the rather strong biases in stimulus repetition probabilities. This suggests a largely implicit form of learning, a hypothesis that needs to be tested rigorously in future work. Further, our behavioral modeling required many trials, precluding an assessment of the dynamics of the bias (i.e., weight) adjustment during the blocks of a given sensory environment. This issue should be addressed in future work, using models capable of learning environmental parameters such as transition probabilities (Yu and Cohen, 2009; Meyniel et al., 2016; Glaze et al., 2018; Hermoso-Mendizabal et al., 2020).

Previous work has characterized neural signals underlying idiosyncratic history biases in contexts where these biases would be maladaptive. Such signals were observed in several brain areas and in different formats. In a continuous spatial working memory task, activity-silent codes in prefrontal cortex during mnemonic periods seem to promote memory reactivations, which mediate serial memory biases (Barbosa et al., 2020). Studies of perceptual forced choice tasks have found signatures of persistent population activity reflecting the previous choice in posterior parietal cortex (Morcos and Harvey, 2016; Hwang et al., 2017; Scott et al., 2017; Urai and Donner, 2022), prefrontal cortex (Mochol et al., 2021), and motor cortex (Pape and Siegel, 2016; Urai and Donner, 2022). Specifically, human MEG work showed that history-dependent modulations of parietal cortical activity in the gamma-band spanned the intervals between trials and mediated idiosyncratic choice repetition biases (Urai and Donner, 2022). Such an effect was not observed for the motor beta-rebound that was similarly sustained into the next decision interval (Urai and Donner, 2022). Importantly, none of these studies quantified the build-up rate of action-selective motor cortical activity on the subsequent trial.

Idiosyncratic history biases are reflected in a persistent baseline state of action-selective neural population activity in monkey prefrontal cortex, during decision formation accompanied by a subtler modulation of the build-up rate (Mochol et al., 2021). Another human MEG study derived an action-independent proxy of the neural DV from sensor-level MEG data that required two successive judgments within a trial (Rollwage et al., 2020). The initial decision biased the subsequent build-up of that DV-proxy in a manner that depended on the consistency of new evidence with the initial decision and the confidence in that initial decision. Critically, no previous study has investigated the flexible and performance-increasing history biases that we have manipulated and studied here.

Our current results resemble the results from block-wise manipulations of the probability of a specific stimulus category (i.e., not of transitions across stimulus categories): this also biases the build-up of saccade-selective activity in monkey posterior parietal cortex (Hanks et al., 2011), just like what we found here for hand-movement selective motor cortical activity in humans, albeit with strong, but lawful, trial-by-trial variations in our current

setting (Fig. 1E). It is tempting to interpret both as downstream expressions of perceptual expectations in cortical circuitry involved in action planning. Indeed, modulating the build-up of an evolving decision variable by prior expectations can be useful in accumulation-to-bound models when reliability of the evidence varies from decision to decision (Hanks et al., 2011; Moran, 2015). Whether or not the neural signatures of idiosyncratic choice history biases studied in previous work have similar cognitive content and underlying mechanisms remains an open question.

Our results indicate that dynamic and adaptive expectations bias the dynamics of neural signatures of action planning during decision formation. How are these expectations implemented in upstream neural populations, so as to yield the selective changes in M1 ramping dynamics observed here? One possibility is that choice history biases the state of sensory cortex (Nienborg and Cumming, 2009; St. John-Saaltink et al., 2016), for example via feedback from cortical areas involved in decision formation (Wimmer et al., 2015). Another possibility is that the expectations shape the read-out of sensory evidence by the evidence accumulator, with preferential accumulation of evidence that matches the expectation, in line with active inference (Friston, 2010). Yet another possibility is that the evidence accumulator receives non-sensory input from brain regions encoding history information in a sustained fashion (Talluri et al., 2021; Urai and Donner, 2022). In all these different schemes, dynamic expectations would need to be constructed in a highly flexible, context-dependent fashion in order to give rise to the adaptive biasing of action-selective activity observed here.

Funding

This work has been supported by the Deutsche Forschungsgemeinschaft (DFG), projects DO1240-4-1, DO1240_2-2, and SFB936/Z3 and by the Federal Ministry of Education and Research (BMBF), projects 01EW2007B and 01EW2007A (all to THD).

Acknowledgements

We thank Niklas Wilming for discussion on MEG source reconstruction and Jaime de la Rocha, Anne Urai, Bharath Chandra Talluri, and Alessandro Toso for discussion and comments on the manuscript.

Author contributions

Conceptualization: A.B., T.H.D.; methodology, software, investigation, formal analysis, data curation, visualization: A.B.; writing: A.B., T.H.D.; supervision: T.H.D.; funding acquisition: T.H.D.; project management: T.H.D.

Materials and Methods

Participants

42 healthy human observers (27 female, 15 male) participated in the study using magnetoencephalography (MEG) together with pupillometry. All participants gave their

written informed consent. The experiment was approved by the local ethical review board (Ärztchamber Hamburg reference number PV4714). Two participants showed performance around chance level in the training session and therefore, did not participate in the MEG sessions. Two more participants were excluded from the analysis so that 38 participants remained for the data analysis. One of the excluded participants did not respond within the response interval on a substantial number of trials (31 percent of trials), and the other participant was excluded due to excessive MEG artifacts. We excluded three single sessions from separate participants due to substantially worse performance than during the rest of the sessions.

Behavioral task

We used a random dot motion discrimination task with varying levels of evidence strength (motion coherence) spanning the psychophysical threshold (Fig. 1A). Participants had to judge whether a cloud of coherently moving signal dots embedded in dynamic noise was either moving upwards or downwards. To interrogate the adaptability of choice history biases, participants performed the task in three different stimulus environments, defined by varying levels of autocorrelation between stimulus categories (upwards or downwards) across trials. In a ‘Neutral’ environment the direction of motion was chosen at random on each trial, in a ‘Repetitive’ environment the previous motion direction was more likely to be repeated (80% repetition probability) and in an ‘Alternating’ environment the previous motion direction was more likely to be alternated (20% repetition probability). The resulting fractions of upward and downward motion stimuli were approximately equal within each environment: The group average frequency of upward trials was 0.502 for Neutral, 0.507 for Repetitive, and 0.500 for Alternating.

The magnetoencephalography data of this experiment allowed for identifying the neural correlates underlying the choice history bias adjustment.

Stimuli

Random dot kinematograms contained 117 white dots at a density of 6 dots/deg² on a gray screen. Each dot had a size of 0.06°. The dots were moving within a circular aperture of 2.5° radius of visual angle centered around a fixation cross of 0.2° x 0.2°. The aperture was placed 3.5° below the center of the screen. Random dots (0% coherence) were presented throughout the whole trial to guarantee constant luminance in order to avoid luminance-induced changes in pupil diameter. During the evidence interval, coherently moving signal dots were superimposed onto the random noise dots. The signal dots moved either upwards or downwards (or in random directions in case of 0% motion coherence). The motion coherence, i.e., the percentage of coherently moving dots, was chosen from trial to trial at random out of 5 levels (0, 3, 9, 27, 81%) under the constraint that each block contained an equal number of trials per motion coherence and direction. The signal dots were moving with a velocity of 11.5°/s and each dot had a lifetime of 10 frames. Three variants of dot motion (at the same coherence and direction) were presented in an interleaved fashion within each trial.

Trial structure

The fixation cross changed its color to indicate different periods within each trial. Each trial started with a fixation interval of 0.75 – 1.5 s (uniformly distributed), during which the fixation cross was colored in red. After the fixation interval, the fixation cross turned green to indicate the onset of coherent motion. After a fixed evidence duration of 0.75 s, the signal dots disappeared from the screen and the fixation cross turned red again to indicate the

start of the response interval. Participants were instructed to report their choice with a left- or right-hand button-press. The choice-hand mapping was counterbalanced within each participant and randomly chosen per block with the restriction that both choice-hand mappings occurred once per stimulus environment per session. After button-press or a maximum response time of 1.25 s in case no response was given, the fixation cross turned blue and the inter-trial interval started. After a uniformly distributed interval of 1.5 – 2.5 s (pupil rebound time after response), participants received auditory feedback (0.15 s) about the accuracy of their response. A high tone (1100 Hz) was given for a correct response, a low tone (150 Hz) for an incorrect response, an intermediate tone (440 Hz) after a 0% coherence trial (accuracy not defined) and a white noise tone if the participant did not respond within the maximum response time. The inter-trial interval continued for another 2 – 2.5 s (uniformly distributed). Participants were instructed to fixate the cross during the entire trial and not to blink during all periods but the inter-trial interval.

Participants performed one training session and three MEG sessions of 2 hours each. Each session consisted of 6 blocks of 99 trials each. The repetition probability between the two motion directions remained constant within each block but randomly varied across blocks under the constraint that each session contained two blocks of each environmental condition. Participants were not informed about the manipulation of the stimulus sequence.

Behavioral modeling of choice history bias

Logistic regression model with history bias

To quantify the influence of the history of previous choices and stimulus categories on the current choice, we used a logistic regression model with a history-dependent bias term that shifted the psychometric function along the horizontal axis (Fründ et al., 2014; Urai et al., 2017; Braun et al., 2018). Specifically, the probability of making one of the two choices $c_t = 1$ ($c_t = 1$ for ‘choice up’, $c_t = -1$ for ‘choice down’) on trial t was described by:

$$P(r_t = 1 | \tilde{s}_t, \mathbf{h}_t) = \gamma + (1 - \gamma - \lambda) g(\delta(\mathbf{h}_t) + \alpha \tilde{s}_t). \quad (1)$$

γ and λ were the lapse rates for the choices $c_t = 1$ and $c_t = -1$, and $g(x) = \frac{1}{1+e^{-x}}$ was the logistic function. $\tilde{s}_t$ was the signed stimulus intensity (i.e., motion coherence times stimulus category; ‘up’ or ‘down’, coded as 1 and -1) and α was the slope of the stimulus-dependent part of the psychometric function, quantifying perceptual sensitivity. The bias term

$$\delta(\mathbf{h}_t) = \delta' + \delta_{hist}(\mathbf{h}_t) = \delta' + \sum_{k=1}^{2n} \omega_k h_{kt} \quad (2)$$

, i.e., the offset of the psychometric function, consisted of an overall bias δ' for one specific choice (‘up’ or ‘down’) and a history-dependent bias term $\delta_{hist}(\mathbf{h}_t) = \sum_{k=1}^{2n} \omega_k h_{kt}$, which was the sum of the preceding n (see Determination of model order below for determination of n) choices c_{t-1} to c_{t-n} and the preceding n stimulus categories z_{t-1} to z_{t-n} , each multiplied with a weighting factor ω_k . The vector $\mathbf{h}_t$ was made up of the last n choices and stimulus categories: $\mathbf{h}_t = (c_{t-1}, \dots, c_{t-n}, z_{t-1}, \dots, z_{t-n})$. Upward and downward choices and stimulus categories were coded as 1 and -1 and stimuli with zero motion coherence were set to 0. The weighting factors ω_k specified the influence of each of the n preceding choices and stimulus categories on the current choice. Positive values of ω_k referred to a tendency to repeat, and negative values of ω_k referred to a tendency to alternate the choice or stimulus category at the corresponding lag. All parameters were fit by maximizing the log-likelihood $L =$

$\sum_t \log(P(r_t = 1|\tilde{s}_t, \mathbf{h}_t))$ using an expectation maximization algorithm (Fründ et al., 2014). The slope was fitted separately for each session and then averaged across sessions.

In [Figure S2A](#), we tested the clustering of vector angles of the shift between the weights from Neutral and the weights from the Repetitive or Alternating environments, respectively and the difference of these shifts between both environments. The same qualitative pattern of results was observed when the shift angles for Repetitive and Alternating environments were computed with respect to the origin rather than the individual data points for Neutral.

In [Fig. 1F](#), we computed an individual measure of bias adjustment as the length of the vector between the weights from Repetitive and Alternating from [Fig. S2A](#).

Determination of model order

To avoid overfitting, we determined the model order, i.e., the number of lags n in the logistic regression model that described the behavioral data best, separately for each subject and each environmental condition using a 6-fold cross-validation procedure. We split the data into six test and training sets. Each test set contained one out of the six blocks of each environment, and the training set contained the remaining five blocks. We shuffled the assignment of the test block and the training blocks across all six possibilities resulting in six different pairs of test and training datasets. For each training dataset, we fitted the logistic regression model with varying number of lags ranging from 0 (no history) to 7 lags. For each fold and model order, we computed the log-likelihood $L = \sum_t \log(P(r_t = 1|\tilde{s}_t, \mathbf{h}_t))$ using the choices and stimuli from the test data and the fitted model parameters, i.e., history weights, general bias, lapse rate and slope from the corresponding training data. We averaged the log-likelihood values for each subject and model order across the six folds. The model with the maximum log-likelihood value defined the best fitting model order n that was used for the subsequent analyses ([Fig. S1](#)). For those subjects for which the model without history bias, i.e., zero lags, was the best fitting model for one biased environment, we set the model order for the corresponding environment to 1 for the behavioral analyses. We excluded two subjects from the analyses of the MEG data for which the model without history bias, i.e., zero lags, was the best fitting model for both biased environments, as those subjects did not adapt their choice behavior to the statistical structure of the environment.

Single-trial bias estimates

To obtain an estimate of the bias at each single trial ([Fig. 1E](#)), we computed the bias term $\delta(h_t) = \delta' + \delta_{hist}(h_t) = \delta' + \sum_{k=1}^{2n} \omega_k h_{kt}$ using the vector of previous choices and stimulus categories $\mathbf{h}_t$ from each test dataset (block) and the general bias δ' and history weights ω_k for the previous choices and stimulus categories at lag $k = 1$ to n fitted from the corresponding training dataset. By fitting the model excluding the block from the test dataset, we guaranteed that the single-trial bias estimates were not contaminated by the data that they were supposed to predict. The sign of the single-trial bias δ determined the tendency for an 'up' (for a positive sign) or 'down' (for a negative sign) choice before stimulus presentation (different from the history weights ω_k , which indicate a tendency to repeat or alternate). The magnitude of the single-trial bias δ defined the strength of this tendency.

We binned the single-trial bias estimates into three bins of equal size separately for each subject. The low bin contained the values in the 0-33% quantile, the medium bin contained the values in the 33-66% quantile and the high bin contained the values in the 66-100% quantile. On average, the values in the low bin were negative corresponding to a bias for choice 'down', the medium bin contained a bias close to zero and the values in the high bin were positive indicating a bias for choice 'up'.

MEG data acquisition and analysis

Data acquisition

MEG data was recorded with a whole-head 275-channel CTF system at a sampling rate of 1200 Hz. We simultaneously recorded saccades and pupil dilation using an EyeLink 1000 Long Range Mount (SR Research, Osgoode, Ontario, Canada) and vertical and horizontal EOG as well as a bipolar electrocardiogram (ECG) using Ag/AgCl electrodes. To monitor the subjects' head position, we used three fiducial coils: one above the nasion and one each in the left and right auricle. We used online head-localization (Stolk et al., 2013) to adjust the subjects' head position before each block to maintain the same head position relative to the MEG sensors across blocks within each session. To obtain the same head position across all three MEG sessions, we located the subjects' head position in the second and third session relative to its position in the first session. Stimuli were shown on a screen with a refresh rate of 60 Hz, at a distance of 65 cm from the subjects' eyes using a beamer with a resolution of 1024 x 768 pixels.

Preprocessing

First, the data was down-sampled to 400 Hz and epoched into single trials from fixation (0.75 s before the evidence interval) to 1.5 s after feedback. Then, we cleaned the data from artifacts via visual inspection as well as through semi-automatic artifact rejection routines using the Fieldtrip Toolbox (Oostenveld et al., 2011). We removed trials in which no response was given within the maximum response interval of 1.25 s after evidence offset and trials with excessive head motion > 6 mm deviation from the first trial (Stolk et al., 2013). We removed line noise around 50, 100 and 150 Hz using a bandstop filter and demeaned and detrended the data. To detect artifacts caused by cars passing by the MEG lab, we low pass filtered the data at 1 Hz, applied a Hilbert transform, z-scored the data and removed trials with large amplitudes and a slow drift of the resulting signal via visual inspection. Muscle bursts and squid jumps were detected via visual inspection after applying a 9th order 110-140 Hz Butterworth filter, a Hilbert transform, and z-scoring. Eye blinks and saccades were identified via visual inspection of the vertical and horizontal EOG channels after applying a 1-15 Hz bandpass filter, a Hilbert transform, and z-scoring the data. Trials with muscle bursts, eye blinks or saccades were removed in case those artifacts occurred before the response. The cleaned data was epoched into stimulus-locked (-0.55 to 1.5 s around evidence onset) and response-locked (-0.5 to 1.5 s around button press) segments.

Source reconstruction

We used linearly constrained minimum variance (LCMV) beamforming (Van Veen et al., 1997) and time-frequency decomposition to reconstruct the local field potentials at the source level. We first reconstructed the cortical surface from each participant's anatomical MRI scan using freesurfer (Dale, 1999; Fischl et al., 1999). In case no MRI scan was available (3 subjects), we used an average subject provided by freesurfer, that was obtained from the average across 40 subjects. Then, we aligned the atlases to the cortical surface. We computed head meshes (boundary element method (BEM) surfaces) using fieldtrip (Oostenveld et al., 2011) and the head shape model using MNE (Gramfort et al., 2014). Next, we created the transformation matrix by co-registering the headlock fiducials to the head model separately for each subject and session. A source space (4096 vertices per hemisphere, recursively subdivided octahedron) was computed for each hemisphere, surfaces were converted to a BEM and the BEM solution was computed using MNE. We baseline-corrected the stimulus and response epochs using a baseline interval from 0.35 to 0.1 s before stimulus onset and computed a data covariance matrix from the stimulus epochs separately for each subject

and session. The leadfield (forward solution) was computed using the subject and session-specific transformation matrix, source space and BEM solution. Finally, the LCMV spatial filters (Van Veen et al., 1997) were constructed for each vertex in each region of interest from the forward solution and the data covariance matrix. As regions of interest we focused on a number of topographically organized visual cortical field maps (Wang et al., 2015) and three regions exhibiting action-selective activity lateralization in functional MRI (de Gee et al., 2017): the hand area of primary motor cortex (M1), the junction of intraparietal sulcus/postcentral sulcus IPS/PostCes, and a part of anterior intraparietal sulcus (aIPS).

Spectral analysis

Single-trial complex time-frequency representations of the source-reconstructed signal were computed with a window length of 400 ms in steps of 25 ms using MNE (Gramfort et al., 2014). For the low frequencies (3-37 Hz in steps of 2 Hz), we used one taper and a frequency smoothing of 5 Hz (2.5 Hz half window). For the high frequencies (37-161 Hz in steps of 4 Hz), we used a multitaper approach (using Morlet wavelets windowed with discrete prolate spheroidal sequences (DPSS)) with seven tapers and a frequency smoothing of 20 Hz (10 Hz half window). Then, the LCMV beamformer weights of the vertices within each region of interest (ROI) were applied to the complex output of the time-frequency representations before computing the power and averaging across trials and vertices. For each ROI and frequency, we computed the baseline as the average power across trials during the interval ranging from 350 to 100 ms before evidence onset, separately for each subject and session. The data for each ROI and frequency was then transformed into percent signal change from the corresponding baseline.

Regions of interest

We delineated power at specific regions of interest that have been shown to be involved in decision-making, the decision-related dynamics of which have been characterized in detail in previous work (Wilming et al., 2020; Murphy et al., 2021). During decision formation, sensory evidence is encoded in visual cortex. This signal is accumulated across time into a decision variable in association cortex and transformed into a motor action in motor cortex (Gold and Shadlen, 2007; Wang, 2008; Siegel et al., 2011). Specifically, we selected the ROIs from the Wang atlas (Wang et al., 2015) and combined them into the following clusters of interest: primary occipital cortex V1, early occipital cortex V2-4, dorsal occipital cortex V3A/B, intraparietal sulcus IPS0/1 and IPS2/3, lateral occipital cortex LO1 and LO2, temporal occipital area MT+ (MT and MST), ventral occipital cortex VO1 and VO2, parahippocampal cortex PCH1 and PCH2. We used the following regions that have previously been identified to show choice-predictive lateralized activity (de Gee et al., 2017): anterior intraparietal sulcus aIPS, intraparietal sulcus/postcentral sulcus IPS/PostCes, hand area of primary motor cortex M1.

Definition of the time window of linear build-up of lateralized activity in M1

To test for a neural correlate of a bias in drift rate, we first determined the time window of the approximately linear build-up of lateralized activity in M1. During evidence accumulation, choice-predictive motor preparatory activity (a lateralized suppression of beta-band power) builds up contra- vs. ipsilateral to the upcoming button-press. This signal has been shown to exhibit the hallmark signatures of evidence accumulation postulated by the drift diffusion model (Donner et al., 2009; de Lange et al., 2013; Pape and Siegel, 2016). Hence, we used this signal as a neural correlate of the accumulated evidence. To determine the time window of evidence accumulation, we fitted a bilinear regression to the slope of the beta-band (12-36 Hz) power contra- vs. ipsilateral to the button-press in M1 pooled across

environmental conditions and averaged across trials (Fig. 4A). We used a time window with a buffer of 250 ms after the intersection point of the fitted lines and 50 ms before the minimum of the beta lateralization to test our hypotheses.

Removal of beta rebound from previous trial

After the motor-response, beta lateralization flips its sign – the so-called beta rebound (Pfurtscheller et al., 1996). This signal leaks into the next trial, which may cause a motor-response alternation bias (Pape and Siegel, 2016). We computed the beta rebound as the beta-band time course contra- vs. ipsilateral to the previous button-press in M1, pooled across environmental conditions and averaged across trials, and normalized it to a unit vector $\mathbf{r}$, separately for each subject. In control analyses for the results from Fig. 4B, C, D and E, we removed the beta rebound from the time course of the beta-band lateralization to isolate the effect of the bias adjustment to the statistical structure of the environment. The residual beta time course $\mathbf{y}^*$ was computed as the difference of the original beta time course $\mathbf{y}$ and its orthogonal projection with the beta rebound:

$$\mathbf{y}^* = \mathbf{y} - (\mathbf{y}^T \mathbf{r}) \mathbf{r}. \quad (3)$$

Assessment of bias-dependent dynamics of action-selective activity

To finally test for a bias-dependent evidence accumulation, we analyzed the beta lateralization conditioned on the behavioral bias at each single trial binned into three bins: a low bin corresponding to a bias for a ‘down’ choice, a medium bin with a bias close to zero and a high bin with a bias for an ‘up’ choice (see section Single-trial bias estimates for details) (Fig. 1E). The single-trial bias shifts the current choice at a given level of evidence. Consequently, the single-trial bias bins correlated with the final choice. The low bin primarily contained trials that resulted in a ‘down’ choice and the high bin primarily contained trials that resulted in an ‘up’ choice. To remove the effect of the final choice to isolate the effect of the single-trial bias, we subsampled the data such that each bias bin contained an equal number of up and down choices separately for each subject. To this end, we randomly drew the number of trials of the inferior choice from the data containing the predominant choice, separately for each bin. We repeated this procedure 1000 times and averaged the data across the draws. We finally computed the time course of the residual beta-band activity of the subsampled data contra- vs. ipsilateral to the button-press for the up choice. Averaging across the low bin with a sign flip and the high bin (without sign flip) yielded the beta lateralization contra- vs. ipsilateral to the button-press that was mapped onto the choice that was in line with the bias (Fig. 4D). We then computed the slope of the build-up of the beta lateralization during the previously defined time window of linear build-up of lateralized activity in M1 (see Definition of the time window of linear build-up of lateralized activity in M1; Fig. 4A) via linear regression (Fig. 4E).

Single-trial regression of history bias and evidence on action-selective activity

We used a linear regression model to quantify the influence of the current sensory evidence (i.e., the signed motion coherence) as well as of the single trial bias on the single-trial modulation of M1 power lateralization during each time point t :

$$beta_lat_t = \beta_0 + \beta_1 * coh + \beta_2 * bias \quad (4)$$

where $beta_lat_t$ was the beta-power lateralization relative to the hand coding up-choices in a given block during time point t , coh was the signed motion coherence and $bias$ was the

single trial bias. The power values for each time point t , frequency f and sensor c were normalized and baseline-corrected via the decibel (dB) transform before computing the beta lateralization: $dB_{t,f,c} = 10 * \log_{10}(power_{t,f,c}/baseline_{f,c})$, where $baseline_{f,c}$ was the trial-averaged power during the baseline interval (350 to 100 ms before onset of coherent motion). All regressors as well as power values were z-scored prior to the regression analysis. We expected a negative influence of the signed motion coherence as well as of the single trial bias on the motor beta lateralization contra- vs. ipsilateral to the button-press for up responses (Fig. 4F, G).

For the analysis of the influence of the single trial bias on the baseline M1 beta lateralization (350 to 100ms before evidence onset; Fig. 3D), we used an analogous regression analysis but without using the signed motion coherence as a regressor because the onset of coherent motion started only after the baseline interval:

$$beta_lat_{baseline} = \beta_0 + \beta_1 * bias \quad (5)$$

Single-trial regression of history bias and evidence on slope of action-selective activity

We used the corresponding regression analysis for the slope of the motor beta lateralization separately for current up and down responses:

$$beta_slope_t = \beta_0 + \beta_1 * coh + \beta_2 * bias \quad (6)$$

To this end we computed the slope of the M1 beta lateralization time course using a sliding window of 200ms. The slope for each time window t as well as the regressors were z-scored before computing the regression. We plotted the beta weights at the center of each 200ms time-window that was used to compute the slope of the beta lateralization (Fig. 4F, G grey line).

Statistical tests

We used parametric two-tailed t tests to test the effect of the previous stimulus category on the shift and the slope of the psychometric function in order to also provide Bayes factors (Bf) (Rouder et al., 2009) (Figure 1C). $Bf_{10} < \frac{1}{3}$ corresponds to evidence in favor of the null hypothesis, $Bf_{10} > 3$ refers to evidence for the alternative hypothesis and $Bf_{10} = 1$ corresponds to inconclusive evidence. We used Pearson correlation for computing the partial correlation between the bias adjustment and performance (Fig. 1F and S3). We used nonparametric permutation tests (Efron and Tibshirani, 1998) with $N = 10000$ permutations to test the previous stimulus weights (Fig. 1D), the baseline state of the beta lateralization (Fig. 3B, C), the slope of the build-up of motor preparatory activity (Fig. 4C, E) as well as for the regression of the single-trial history bias on action-selective activity during the baseline interval (Fig. 3D). Cluster-based permutation tests were used for time-frequency responses (Fig. 2, 3A) and for time courses of beta-band power (Fig. 4). We used circular statistics (Rayleigh's test) to test the clustering of vector angles between the origin and the weights from the Neutral environment as well as between the weights from Neutral and the weights from the Repetitive or Alternating environments, respectively (Figure S2A). To test the difference in mean directions of adjustment between the Repetitive and the Alternating environment, we used a Hotelling test (van den Brink et al., 2014) (Figure S2A).

Supplementary information

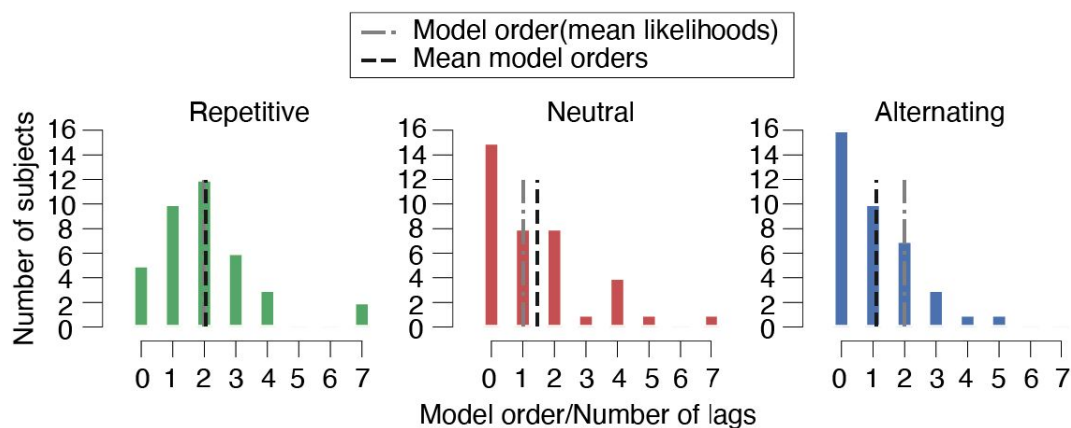


Figure S1.

Best-fitting model orders for behavioral history bias.

Best-fitting model orders, defined as number of lags n of the history bias terms of the logistic regression model (Materials and Methods) were determined via a cross-validation procedure, separately for each individual and for the three different environments. Shown are histograms of the resulting model orders across subjects, separated by sensory environment. Dashed vertical lines, mean of model orders (black) and model order determined from mean of likelihoods (grey).

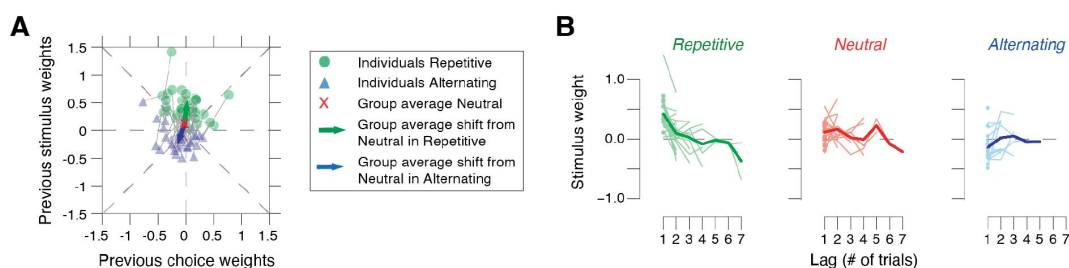


Figure S2.

Patterns of individual choice history biases across environments.

A. Impact of previous choices versus impact of previous stimulus categories on current choice for lag 1. Green dots and blue triangles refer to values from individual participants in Repetitive and Alternating, respectively. Grey lines connect values from both environments. Green and blue arrows indicate shift of group averages from Neutral (red x) during Repetitive and Alternating, respectively. Positive weights corresponded to tendency to repeat, and negative weights to tendency to alternate previous choice (x) or stimulus category (y). Angles of vectors from the origin to individual data points differed from uniform in Neutral ($z = 7.536$, $p = 0.0004$; Rayleigh's test). Individual vector angles of shift from Neutral differed from uniform in Repetitive ($z = 19.382$, $p < 0.0001$) and Alternating ($z = 21.287$, $p < 0.0001$; Rayleigh's test), with a difference in shift between both environments ($F_{(2,34)} = 75.79$, $p < 0.0001$, Hotelling test). **B.** History kernels quantifying the impact of previous stimulus categories on current choice as a function of lag, for the three environments. Circles and thin lines, individual subjects; thick lines, group average. Circles are drawn for subjects whose best-fitting lag is 1. The predominant effect of sensory environment is evident at lag 1.

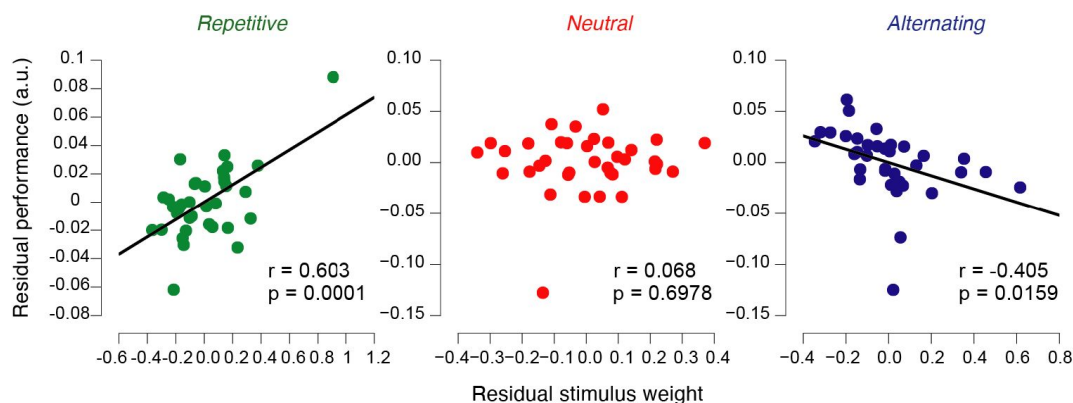


Figure S3.

Performance in biased environments depends on strength of previous stimulus weights.

Partial correlation analysis between previous stimulus weights and individual performance in each environment after factoring out the correlation of both variables with perceptual sensitivity (see main text, Materials and Methods). Data points are the residuals from two separate regressions: previous stimulus weight on sensitivity (x-axis) and sensitivity on proportion correct (y-axis).

References

- Abrahamyan A , Silva LL , Dakin SC , Carandini M , Gardner JL (2016) **Adaptable history biases in human perceptual decisions** *Proceedings of the National Academy of Sciences* **113**:
- Akaishi R , Umeda K , Nagase A , Sakai K (2014) **Autonomous Mechanism of Internal Choice Estimate Underlies Decision Inertia** *Neuron* **81**:195–206
- Bogacz R , Brown E , Moehlis J , Holmes P , Cohen JD (2006) **The physics of optimal decision making: A formal analysis of models of performance in two-alternative forced-choice tasks** *Psychological Review* **113**:700–765
- Braun A , Urai AE , Donner TH (2018) **Adaptive History Biases Result from Confidence-weighted Accumulation of Past Choices** *The Journal of Neuroscience* 2189–17
- Brody CD , Hanks TD (2016) **Neural underpinnings of the evidence accumulator** *Current Opinion in Neurobiology* **37**:149–157
- Brunton BW , Botvinick MM , Brody CD (2013) **Rats and Humans Can Optimally Accumulate Evidence for Decision-Making** *Science* **340**:95–98

- Busse L , Ayaz A , Dhruv NT , Katzner S , Saleem AB , Scholvinck ML , Zaharia AD , Carandini M (2011) **The Detection of Visual Contrast in the Behaving Mouse** *Journal of Neuroscience* **31**:11351–11361
- Dale AM (1999) **Optimal experimental design for event-related fMRI** *Human Brain Mapping* **8**:109–114
- de Gee JW , Colizoli O , Kloosterman NA , Knapen T , Nieuwenhuis S , Donner TH (2017) **Dynamic modulation of decision biases by brainstem arousal systems** *eLife* **6**
- de Lange FP , Rahnev DA , Donner TH , Lau H (2013) **Prestimulus Oscillatory Activity over Motor Cortex Reflects Perceptual Expectations** *Journal of Neuroscience* **33**:1400–1410
- Donner TH , Siegel M , Fries P , Engel AK (2009) **Buildup of Choice-Predictive Activity in Human Motor Cortex during Perceptual Decision Making** *Current Biology* **19**:1581–1585
- Efron B , Tibshirani R (1998) **An introduction to the bootstrap**
- Fernberger SW (1920) **Interdependence of judgments within the series for the method of constant stimuli** *Journal of Experimental Psychology* **3**:126–150
- Fischer AG , Nigbur R , Klein TA , Danielmeier C , Ullsperger M (2018) **Cortical beta power reflects decision dynamics and uncovers multiple facets of post-error adaptation** *Nat Commun* **9**:5038
- Fischer J , Whitney D (2014) **Serial dependence in visual perception** *Nature Neuroscience* **17**:738–743
- Fischl B , Sereno MI , Dale AM (1999) **Cortical Surface-Based Analysis** *NeuroImage* **9**:195–207
- Friston K (2010) **The free-energy principle: a unified brain theory?** *Nature Reviews Neuroscience* **11**:127–138
- Fritsche M , Mostert P , de Lange FP (2017) **Opposite Effects of Recent History on Perception and Decision** *Current Biology* **27**:590–595
- Fründ I , Wichmann FA , Macke JH (2014) **Quantifying the effect of intertrial dependence on perceptual decisions** *Journal of Vision* **14**:9–9
- Glaze CM , Filipowicz ALS , Kable JW , Balasubramanian V , Gold JI (2018) **A bias-variance trade-off governs individual differences in on-line learning in an unpredictable environment** *Nat Hum Behav* **2**:213–224
- Glaze CM , Kable JW , Gold JI (2015) **Normative evidence accumulation in unpredictable environments** *eLife* **4**

- Gold JI , Law C-T , Connolly P , Bennur S (2008) **The Relative Influences of Priors and Sensory Evidence on an Oculomotor Decision Variable During Perceptual Learning** *Journal of Neurophysiology* **100**:2653–2668
- Gold JI , Shadlen MN (2007) **The Neural Basis of Decision Making** *Annual Review of Neuroscience* **30**:535–574
- Gramfort A , Luessi M , Larson E , Engemann DA , Strohmeier D , Brodbeck C , Parkkonen L , Hämäläinen MS (2014) **MNE software for processing MEG and EEG data** *NeuroImage* **86**:446–460
- Hanks TD , Kopec CD , Brunton BW , Duan CA , Erlich JC , Brody CD (2015) **Distinct relationships of parietal and prefrontal cortices to evidence accumulation** *Nature* **520**:220–223
- Hanks TD , Mazurek ME , Kiani R , Hopp E , Shadlen MN (2011) **Elapsed Decision Time Affects the Weighting of Prior Probability in a Perceptual Decision Task** *Journal of Neuroscience* **31**:6339–6352
- Hermoso-Mendizabal A , Hyafil A , Rueda-Orozco PE , Jaramillo S , Robbe D , de la Rocha J (2020) **Response outcomes gate the impact of expectations on perceptual decisions** *Nat Commun* **11**:1057
- Hwang EJ , Dahlen JE , Mukundan M , Komiyama T (2017) **History-based action selection bias in posterior parietal cortex** *Nature Communications* **8**
- Kim TD , Kabir M , Gold JI (2017) **Coupled Decision Processes Update and Maintain Saccadic Priors in a Dynamic Environment** *The Journal of Neuroscience* **37**:3632–3645
- Little S , Bonaiuto J , Barnes G , Bestmann S , Ganguly K (2019) **Human motor cortical beta bursts relate to movement planning and response errors** *PLoS Biol* **17**:
- Meyniel F , Maheu M , Dehaene S , Gershman SJ (2016) **Human Inferences about Sequences: A Minimal Transition Probability Model** *PLOS Computational Biology* **12**:
- Mobbs D , Trimmer PC , Blumstein DT , Dayan P (2018) **Foraging for foundations in decision neuroscience: insights from ethology** *Nat Rev Neurosci* **19**:419–427
- Mochol G , Kiani R , Moreno-Bote R (2021) **Prefrontal cortex represents heuristics that shape choice bias and its integration into future behavior** *Current Biology* **31**:1234–1244
- Moran R (2015) **Optimal decision making in heterogeneous and biased environments** *Psychonomic Bulletin & Review* **22**:38–53
- Morcos AS , Harvey CD (2016) **History-dependent variability in population dynamics during evidence accumulation in cortex** *Nature Neuroscience* **19**:1672–1681

Murphy PR , Wilming N , Hernandez-Bocanegra DC , Prat-Ortega G , Donner TH (2021) **Adaptive circuit dynamics across human cortex during evidence accumulation in changing environments** *Nat Neurosci* **24**:987–997

Nienborg H , Cumming BG (2009) **Decision-related activity in sensory neurons reflects more than a neuron's causal effect** *Nature* **459**:89–92

O'Connell RG , Dockree PM , Kelly SP (2012) **A supramodal accumulation-to-bound signal that determines perceptual decisions in humans** *Nature Neuroscience* **15**:1729–1735

O'Connell RG , Kelly SP (2021) **Neurophysiology of Human Perceptual Decision-Making** *Annu Rev Neurosci* **44**:495–516

Oostenveld R , Fries P , Maris E , Schoffelen J-M (2011) **FieldTrip: Open Source Software for Advanced Analysis of MEG, EEG, and Invasive Electrophysiological Data** *Computational Intelligence and Neuroscience* **2011**:1–9

Ossmy O , Moran R , Pfeffer T , Tsetsos K , Usher M , Donner TH (2013) **The Timescale of Perceptual Evidence Integration Can Be Adapted to the Environment** *Current Biology* **23**:981–986

Pape A-A , Siegel M (2016) **Motor cortex activity predicts response alternation during sensorimotor decisions** *Nature Communications* **7**:13098

Peixoto D , Verheijn JR , Kiani R , Kao JC , Nuyujukian P , Chandrasekaran C , Brown J , Fong S , Ryu SI , Shenoy KV , Newsome WT (2021) **Decoding and perturbing decision states in real time** *Nature* **591**:604–609

Pfurtscheller G , Stancák A , Neuper C (1996) **Post-movement beta synchronization. A correlate of an idling motor area?** *Electroencephalography and Clinical Neurophysiology* **98**:281–293

Ratcliff R , McKoon G (2008) **The Diffusion Decision Model: Theory and Data for Two-Choice Decision Tasks** *Neural Computation* **20**:873–922

Rollwage M , Loosen A , Hauser TU , Moran R , Dolan RJ , Fleming SM (2020) **Confidence drives a neural confirmation bias** *Nat Commun* **11**:2634

Rouder JN , Speckman PL , Sun D , Morey RD , Iverson G (2009) **Bayesian t tests for accepting and rejecting the null hypothesis** *Psychonomic Bulletin & Review* **16**:225–237

Scott BB , Constantinople CM , Akrami A , Hanks TD , Brody CD , Tank DW (2017) **Fronto-parietal Cortical Circuits Encode Accumulated Evidence with a Diversity of Timescales** *Neuron* **95**:385–398

Shadlen MN , Kiani R (2013) **Decision Making as a Window on Cognition** *Neuron* **80**:791–806

- Sherman MA , Lee S , Law R , Haegens S , Thorn CA , Hämäläinen MS , Moore CI , Jones SR (2016) **Neural mechanisms of transient neocortical beta rhythms: Converging evidence from humans, computational modeling, monkeys, and mice** *Proc Natl Acad Sci USA* 113
- Siegel M , Donner TH , Oostenveld R , Fries P , Engel AK (2007) **High-Frequency Activity in Human Visual Cortex Is Modulated by Visual Motion Strength** *Cerebral Cortex* 17:732–741
- Siegel M , Engel AK , Donner TH (2011) **Cortical Network Dynamics of Perceptual Decision-Making in the Human Brain** *Frontiers in Human Neuroscience* 5
- St. John-Saaltink E , Kok P , Lau HC , de Lange FP (2016) **Serial Dependence in Perceptual Decisions Is Reflected in Activity Patterns in Primary Visual Cortex** *Journal of Neuroscience* 36:6186–6192
- Stolk A , Todorovic A , Schoffelen J-M , Oostenveld R (2013) **Online and offline tools for head movement compensation in MEG** *NeuroImage* 68:39–48
- Talluri BC , Braun A , Donner TH (2021) **Decision making: How the past guides the future in frontal cortex** *Current Biology* 31:
- Urai AE , Braun A , Donner TH (2017) **Pupil-linked arousal is driven by decision uncertainty and alters serial choice bias** *Nature Communications* 8:14637
- Urai AE , de Gee JW , Tsetsos K , Donner TH (2019) **Choice history biases subsequent evidence accumulation** *eLife* 8
- Urai AE , Donner TH (2022) **Persistent activity in human parietal cortex mediates perceptual choice repetition bias** *Nat Commun* 13:6015
- van den Brink RL , Wynn SC , Nieuwenhuis S (2014) **Post-Error Slowing as a Consequence of Disturbed Low-Frequency Oscillatory Phase Entrainment** *Journal of Neuroscience* 34:11096–11105
- Van Veen BD , Van Drongelen W , Yuchtman M , Suzuki A (1997) **Localization of brain electrical activity via linearly constrained minimum variance spatial filtering** *IEEE Transactions on Biomedical Engineering* 44:867–880
- Wang L , Mruczek REB , Arcaro MJ , Kastner S (2015) **Probabilistic Maps of Visual Topography in Human Cortex** *Cerebral Cortex* 25:3911–3931
- Wang X-J (2008) **Decision Making in Recurrent Neuronal Circuits** *Neuron* 60:215–234
- Wilming N , Murphy PR , Meyniel F , Donner TH (2020) **Large-scale dynamics of perceptual decision information across human cortex** *Nat Commun* 11:5109
- Wimmer K , Compte A , Roxin A , Peixoto D , Renart A , de la Rocha J (2015) **Sensory integration dynamics in a hierarchical network explains choice probabilities in cortical area MT** *Nature Communications* 6

Wyart V , de Gardelle V , Scholl J , Summerfield C (2012) **Rhythmic Fluctuations in Evidence Accumulation during Decision Making in the Human Brain** *Neuron* **76**:847–858

Yu A , Cohen JD (2009) **Sequential effects: Superstition or rational behavior** *Advances in neural information processing systems* **21**:1873–1880

Author information

A. Braun

Section Computational Cognitive Neuroscience, Department of Neurophysiology and Pathophysiology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany, Department of Psychiatry, Charité Universitätsmedizin, Berlin, Germany

For correspondence: anke.braun86@gmail.com

ORCID iD: [0000-0002-1946-7765](https://orcid.org/0000-0002-1946-7765)

T.H. Donner

Section Computational Cognitive Neuroscience, Department of Neurophysiology and Pathophysiology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany, Bernstein Center for Computational Neuroscience, Charité Universitätsmedizin, Berlin, Germany

For correspondence: anke.braun86@gmail.com

ORCID iD: [0000-0002-7559-6019](https://orcid.org/0000-0002-7559-6019)

Editors

Reviewing Editor

Valentin Wyart

Inserm, France

Senior Editor

Michael Frank

Brown University, United States of America

Reviewer #1 (Public Review):

This paper aims to study the effects of choice history on action-selective beta band signals in human MEG data during a sensory evidence accumulation task. It does so by placing participants in three different stochastic environments, where the outcome of each trial is either random, likely to repeat, or likely to alternate across trials. The authors provide good behavioural evidence that subjects have learnt these statistics (even though they are not explicitly told about them) and that they influence their decision-making, especially on the most difficult trials (low motion coherence). They then show that the primary effect of choice history on lateralised beta-band activity, which is well-established to be linked to evidence accumulation processes in decision-making, is on the slope of evidence accumulation rather than on the baseline level of lateralised beta.

The strengths of the paper are that it is: (i) very well analysed, with compelling evidence in support of its primary conclusions; (ii) a well-designed study, allowing the authors to

investigate the effects of choice history in different stochastic environments.

There are no major weaknesses to the study. On the other hand, investigating the effects of choice/outcome history on evidence integration is a fairly well-established problem in the field. As such, I think that this provides a valuable contribution to the field, rather than being a landmark study that will transform our understanding of the problem.

The authors have achieved their primary aims and I think that the results support their main conclusions. One outstanding question in the analysis is the extent to which the source-reconstructed patches in Figure 2 are truly independent of one another (as often there is 'leakage' from one source location into another, and many of the different ROIs have quite similar overall patterns of synchronisation/desynchronisation.). A possible way to investigate this further would be to explore the correlation structure of the LCMV beamformer weights for these different patches, to ask how similar/dissimilar the spatial filters are for the different reconstructed patches.

Reviewer #2 (Public Review):

In this work, the authors use computational modeling and human neurophysiology (MEG) to uncover behavioral and neural signatures of choice history biases during sequential perceptual decision-making. In line with previous work, they see neural signatures reflecting choice planning during perceptual evidence accumulation in motor-related regions, and further show that the rate of accumulation responds to structured, predictable environments suggesting that statistical learning of environment structure in decision-making can adaptively bias the rate of perceptual evidence accumulation via neural signatures of action planning. The data and evidence show subtle but clear effects, and are consistent with a large body of work on decision-making and action planning.

Overall, the authors achieved what they set out to do in this nice study, and the results, while somewhat subtle in places, support the main conclusions. This work will have impact within the fields of decision-making and motor planning, linking statistical learning of structured sequential effects in sense data to evidence accumulation and action planning.

Strengths:

- The study is elegantly designed, and the methods are clear and generally state-of-the-art
- The background leading up to the study is well described, and the study itself conjoins two bodies of work - the dynamics of action-planning processes during perceptual evidence accumulation, and the statistical learning of sequential structure in incoming sense data
- Careful analyses effectively deal with potential confounds (e.g., baseline beta biases)

Weaknesses:

- Much of the study is primarily a verification of what was expected based on previous behavioral work, with the main difference (if I'm not mistaken) being that subjects learn actual latent structure rather than expressing sequential biases in uniform random environments. Whether this difference - between learning true structure or superstitiously applying it when it's not there - is significant at the behavioral or neural level is unclear. Did the authors have a hypothesis about this distinction? If the distinction is not relevant, is the main contribution here the neural effect?
- The key effects (Figure 4) are among the more statistically on-the-cusp effects in the paper, and the Alternating group in 4C did not reliably go in the expected direction. This is not a huge problem per se, but does make the key result seem less reliable given the clear reliability of the behavioral results
- The treatment of "awareness" of task structure in the study (via informal interviews in only a sub-sample of subjects) is wanting

Reviewer #3 (Public Review):

This study examines how the correlation structure of a perceptual decision making task influences history biases in responding. By manipulating whether stimuli were more likely to be repetitive or alternating, they found evidence from both behavior and a neural signal of decision formation that history biases are flexibly adapted to the environment. On the whole, these findings are supported across an impressive range of detailed behavioral and neural analyses. The methods and data from this study will likely be of interest to cognitive neuroscience and psychology researchers. The results provide new insights into the mechanisms of perceptual decision making.

The behavioral analyses are thorough and convincing, supported by a large number of experimental trials (~600 in each of 3 environmental contexts) in 38 participants. The psychometric curves provide clear evidence of adaptive history biases. The paper then goes on to model the effect of history biases at the single trial level, using an elegant cross-validation approach to perform model selection and fitting. The results support the idea that, with trial-by-trial accuracy feedback, the participants adjusted their history biases due to the previous stimulus category, depending on the task structure in a way that contributed to performance.

The paper then examines MEG signatures of decision formation, to try to identify neural signatures of these adaptive biases. Looking specifically at motor beta lateralization, they found no evidence that starting-level bias due to the previous trial differed depending on the task context. This suggests that the adaptive bias unfolds in the dynamic part of the decision process, rather than reflecting a starting level bias. The paper goes on to look at lateralization relative to the chosen hand as a proxy for a decision variable (DV), whose slope is shown to be influenced by these adaptive biases.

This analysis of the buildup of action-selective motor cortical activity would be easier to interpret if its connection with the DV was more explicitly stated. The motor beta is lateralized relative to the chosen hand, as opposed to the correct response which might often be the case. It is therefore not obvious how the DV behaves in correct and error trials, which are combined together here for many of the analyses.